



**Maryland Comprehensive
Assessment Program**

**Grade 5
MISA
Practice Test**



Maryland
STATE DEPARTMENT OF EDUCATION

Section 1

Directions:

Today, you will take Section 1 of the MISA Practice Test.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the lined space provided in your test book. Be sure to keep your response within the provided space. Only responses written within the lined box provided will be scored.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section ONLY. Do not go past the stop sign.

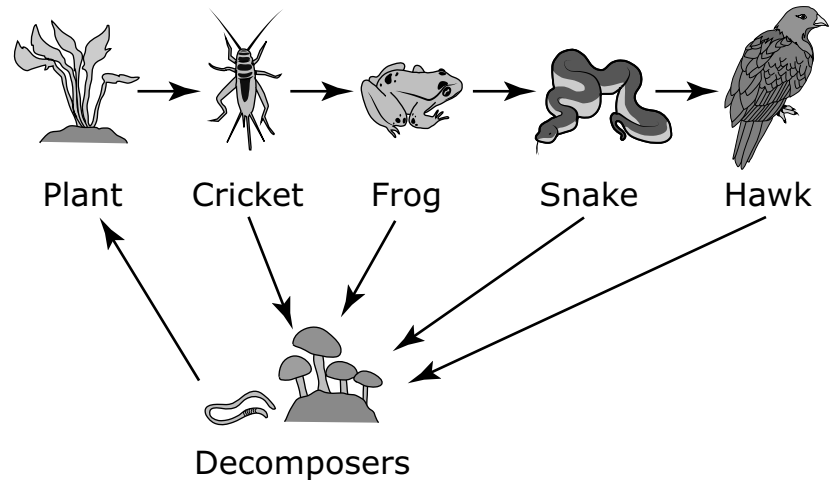
After observing a flower garden outside the school, a science class discussed what plants need to grow and set up an investigation with the help of the teacher. The students added compost, organic matter used to help plants grow, to the soil. They then planted the seeds of five different plants. The students made sure the soil was watered each day. After a few weeks, the plants started to sprout, and the students observed the growth of each plant. A diagram of the school garden is shown.

The row consists of five panels, each with a gray background. The first panel shows three individual rose buds. The second panel shows three leaves with small stems. The third panel shows two small plants with leaves and buds. The fourth panel shows a large, dense cluster of leaves. The fifth panel shows several small flowers with leaves.

Once a week, the students recorded their observations about the garden in their journals. When the plants were fully grown, the students took the plants out of the soil, rinsed them off with water, and had a garden day. During garden day, other students from the school were invited to view the fully grown plants.

After the plants started to grow, the students noticed organisms that were not there before the garden was planted. The students learned that over time the garden had become a small ecosystem. To demonstrate the flow of energy in the garden ecosystem, the students drew a diagram and included decomposers found in the compost, producers, and consumers as shown.

ENERGY FLOW IN A GARDEN ECOSYSTEM





After the lettuce sprouted, the teacher took the lettuce to the school garden and planted it so that the lettuce plant could continue to grow until it was ready to be picked. The students then recorded in their journals the materials needed to recycle lettuce plants and how the lettuce plants can be recycled.

The plant matter was most likely placed in the compost to be

- 2 A student observed a caterpillar eating a leaf in the garden.**

- Ⓐ the plant as a producer.
- Ⓑ the cricket as a producer.
- Ⓒ the plant as a consumer.
- Ⓓ the cricket as a consumer.

- 3 The students included decomposers in the garden ecosystem diagram to show that decomposers have important roles in the garden ecosystem.**

Identify the roles of decomposers.

Select two.

- Ⓐ eat plant roots
- Ⓑ compete with plants for soil
- Ⓒ remove nutrients from the soil
- Ⓓ recycle material back into the soil
- Ⓔ break down plant and animal remains

- 4 The lettuce plant grew new leaves after the teacher placed it in the bowl of water in the classroom.**

The students claimed the plant grew new leaves because the plant

- Ⓐ can survive and grow anywhere.
- Ⓑ was away from insects and other animals in the garden.
- Ⓒ needed a different temperature than the outside garden.
- Ⓓ received the materials it needed from water, air, and sunlight.

Which statement provides evidence that supports this claim?

- Ⓐ The lettuce grew in the garden.
- Ⓑ The lettuce in the classroom sprouted new leaves.
- Ⓒ The students watered the lettuce in the classroom.
- Ⓓ The students saw insects and other animals in the garden.

- Write your response on the lines provided.**

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

While investigating the different forms of energy during science class, a group of students constructed a circuit using a battery, switch, and wires connected to a light bulb in order to cause the light bulb to light up. This led the students to pose questions as to whether a similar design could be used to produce different forms of energy to move objects, make sound, or give off heat.

A diagram of a simple electric circuit. It includes a battery, a motor, a switch, and a propeller. Wires connect the battery to the motor and the switch. The propeller is attached to the motor. Labels with arrows point to each component: Battery, Motor, Propeller, Switch, and Wire.

12

GO ON ►

- 7 In their journals, the students described the transfer of energy needed to cause the propeller to spin.

The best description of the transfer of energy needed to make the toy car move is

- Ⓐ heat energy is converted to sound energy.
- Ⓑ electrical energy is converted to light energy.
- Ⓒ heat energy is converted to the energy of motion.
- Ⓓ electrical energy is converted to the energy of motion.

- 8 Select the three statements that best identify the role of the battery, motor, and switch in the circuit.

- Ⓐ The battery provides the energy of motion to the propeller.
- Ⓑ The battery provides the energy to the circuit.
- Ⓒ The motor provides the energy to the circuit.
- Ⓓ The motor provides the energy of motion to the propeller.
- Ⓔ The switch provides the energy to the circuit.
- Ⓕ The switch opens and closes the circuit.

- 9 Which observation about the final designed toy car provides evidence that the switch controls the flow of the electrical current?

- Ⓐ The car does not move when the switch is open.
- Ⓑ The car does not move when the switch is closed.
- Ⓒ The shape of the switch handle does not affect how far the car moves.
- Ⓓ The length of the switch handle does not affect how far the car moves.

The results will most likely show that the toy car moved as a result of

- 11 Which statement provides the best evidence that energy is being transferred in the toy car investigation?**

- Ⓐ The propeller did not move.
- Ⓑ The students saw light.
- Ⓒ The switch was open.
- Ⓓ The car moved.

- Write your response on the lines provided.**

[illegible]

Section 2

Directions:

Today, you will take Section 2 of the MISA Practice Test.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the lined space provided in your test book. Be sure to keep your response within the provided space. Only responses written within the lined box provided will be scored.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section ONLY. Do not go past the stop sign.

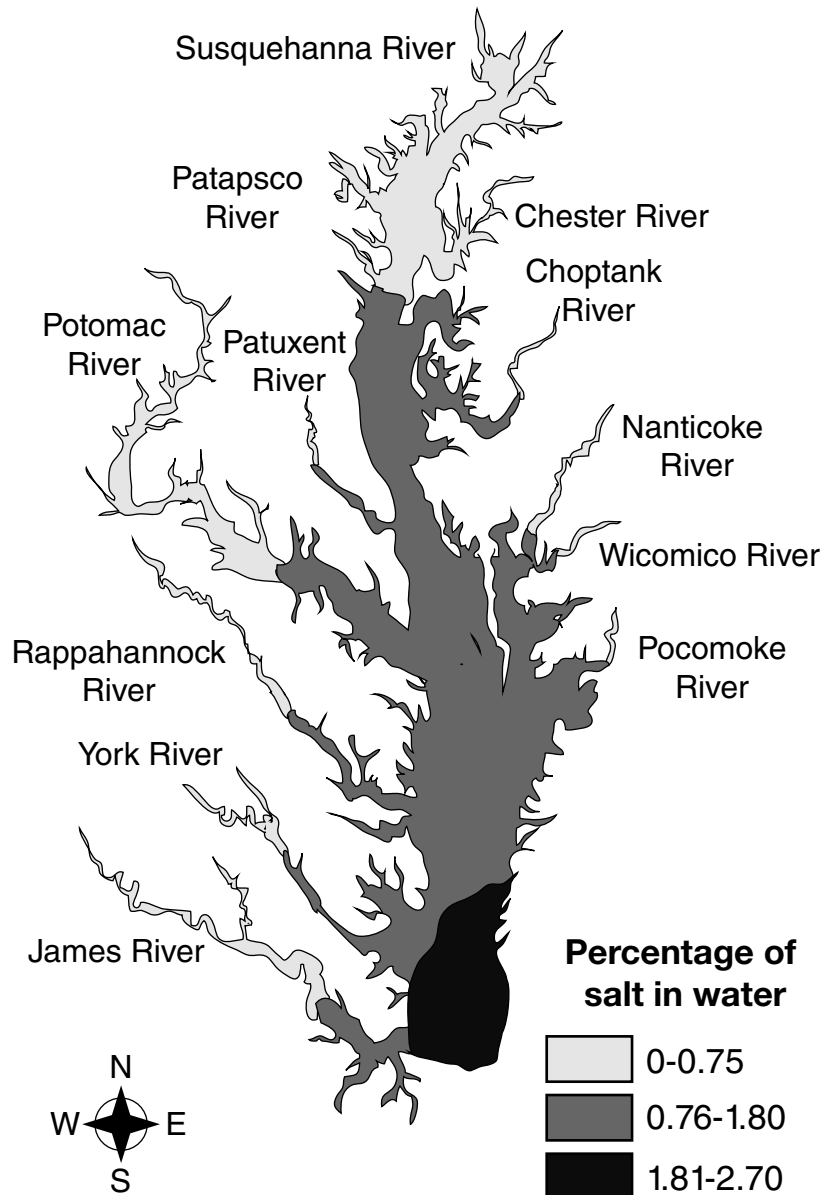
While completing research for the school science fair, a student found the following information on Earth's four major systems. These systems, the hydrosphere, geosphere, atmosphere, and biosphere, are so closely connected that a change in one sphere often results in a change in one or more of the other spheres. The student started the research by looking at a table on the amount of fresh water on Earth, shown as follows.

Glaciers and Ice Caps	Groundwater	Rivers, Lakes, and Other
69%	30%	1%

Fresh water is only 3% of the water found on Earth. The remaining 97% of water on Earth is salt water. The salt water on Earth contains varying concentrations of salt.

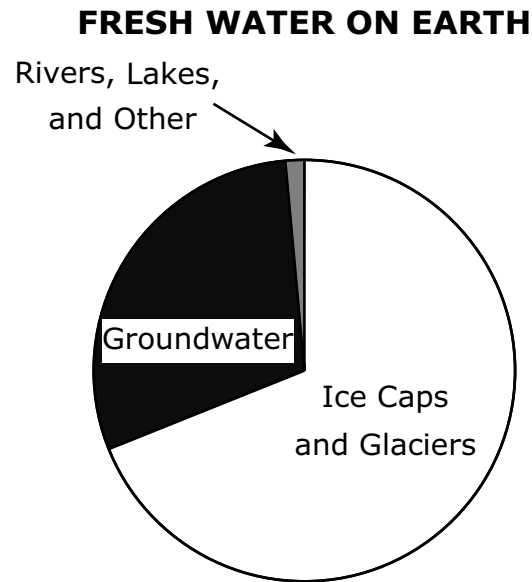
The student decided to continue to research and develop a model of how these systems interact in the Chesapeake Bay watershed. The student discovered that the percentage of salt is the greatest at the mouth of the bay, where the salty ocean waters enter the bay. The student's model of the percentage of salt in the Chesapeake Bay watershed is shown.

CHESAPEAKE BAY WATERSHED



Some organisms have adapted to living in salt water, and some in fresh water. For example, blue crabs can survive in a wide range of salt concentrations; however, during times when there is a greater amount of salt entering the bay, female crabs tend to migrate toward the higher-salinity waters near the mouth of the watershed. Other bay organisms such as water stargrass grow best in fresh water, while eelgrass grows in saltier areas.

- 1 The student used the data to construct a circle graph on the amount of fresh water on Earth, shown below.



The student would most likely use the graph as evidence to support an argument that the greatest amount of Earth's fresh water is

- Ⓐ found in lakes.
- Ⓑ found in rivers.
- Ⓒ trapped in glaciers.
- Ⓓ buried underground.

4 Part A

The student's research provides evidence that the primary sources of salt in the Chesapeake Bay watershed are from a combination of the following two systems:

- Ⓐ biosphere and geosphere
- Ⓑ atmosphere and biosphere
- Ⓒ geosphere and hydrosphere
- Ⓓ hydrosphere and atmosphere

Part B

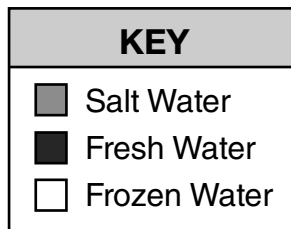
The statement that best describes an interaction between these two systems is that:

- Ⓐ Water vapor in the atmosphere produces salt in the hydrosphere.
- Ⓑ Evaporation from the hydrosphere causes increased salt in the atmosphere.
- Ⓒ Volcanoes in the geosphere cause a decrease of salt for organisms in the biosphere.
- Ⓓ Salt in runoff from the geosphere increases the salt concentration in the hydrosphere.



GO ON TO NEXT PAGE

100



Write your response on the lines provided.

[illegible]

An after-school science club planned to hold a competition that required the use of certain materials to design a small plastic toy van that would travel the farthest after being launched from a starting point. During the competition, club members would launch the vans using a large wooden block on which a magnet was attached. After a van was launched, it could not be touched until it came to a stop.

To prepare for the competition, investigations using the following materials were performed to determine the most effective final investigation design.

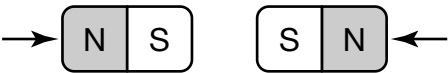
- **1 small plastic toy van**
- **1 launcher magnet**
- **4 labeled bar magnets (W, X, Y, Z)**
- **1 meterstick**
- **1 wooden block**

The launcher magnet could be secured to the wooden block to make a launcher.

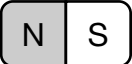
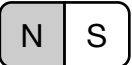
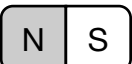
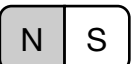
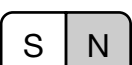
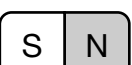
Investigation 1:

Two magnets were oriented such that their south poles faced each other. Then the magnets were pushed together, as shown in the “Magnet Push” diagram. The club members tested other orientations in which different poles faced each other. All orientations were tested, and the observations were recorded in the following table.

MAGNET PUSH



OBSERVATIONS

Magnet Orientations		Observation
		Repelled
		Repelled
		Attracted
		Attracted

The launcher magnet was placed at the 0 centimeter (cm) mark of a meterstick. Each of the labeled magnets was slowly moved from the opposite end of the meterstick toward the launcher magnet, as shown in the diagram.

The diagram shows a horizontal setup on a table. On the left, a bar magnet with its North (N) pole facing right is held in place by a 'Launcher Magnet' indicated by a downward arrow. On the right, another bar magnet with its South (S) pole facing left is being pushed towards the first magnet by a hand. This second magnet is labeled 'Magnet X' with a downward arrow. Below the magnets is a 'Meterstick' with a scale from 0 to 9 cm. The North pole of the first magnet is aligned with the 0 cm mark, and the South pole of the second magnet is aligned with the 4 cm mark.

A club member recorded the distance between the two magnets as soon as the launcher magnet began to move. This procedure was repeated with each magnet, and the results were recorded in the following table.

Labeled Magnet Tested	Distance Between Magnets (cm)
W	4
X	5
Y	8
Z	7

Investigation 3:

After identifying the strongest labeled magnet to attach to the van and the best orientations of the launcher magnet and labeled magnet, the club members attached the labeled magnet to different positions on top of the van. The rear of the van was held at the 0 cm mark of a meterstick. The club members then released the van and recorded how far it traveled for each labeled magnet position. The investigation setup and results for each trial were recorded in the following diagrams.

The diagram illustrates the experimental setup. A toy van is positioned on a horizontal surface, with a meterstick placed below it for measurement. The van's front is aligned with the 0 cm mark on the meterstick. A wooden block is placed behind the van, and a launcher magnet (N-S) and a labeled magnet (S-N) are stacked on top of it. Arrows point from the labels to their respective components: 'Wooden Block' to the block, 'Launcher Magnet' to the N-S magnet, 'Labeled Magnet' to the S-N magnet, and 'Toy Van' to the vehicle. The meterstick is labeled 'Meterstick' and 'cm'.

A diagram showing a car on a horizontal surface. A large grey rectangular magnet is on the left. Above the car's roof are two pairs of magnets. The first pair, positioned over the rear, consists of an 'N' pole (left) and an 'S' pole (right). The second pair, positioned over the front, consists of an 'S' pole (left) and an 'N' pole (right). Below the car is a ruler with markings from 0 to 9 cm.

- 7 The club members were planning an investigation to collect data that would help them win the competition.**

Which two steps should the club members use to collect the most accurate data?

- Ⓐ Make sure the launcher does not move.
- Ⓑ Replace the meterstick with two shorter rulers.
- Ⓒ Use the same van for all trials of the investigation.
- Ⓓ Change two variables after each trial of the investigation.
- Ⓔ Place the van so that the front of the van faces the wooden block.

- 8 Which question was most likely answered in Investigation 2?**

- Ⓐ Which magnet slid most easily across the table?
- Ⓑ Which magnet had the strongest magnetic force?
- Ⓒ How far apart did the magnets need to be held to keep them from attracting each other?
- Ⓓ How was the distance between the magnets affected by the orientations of the magnets?



-

-

11 Which variable in Investigation 3 can be changed for each trial in order for the procedure to remain a fair test?

- Ⓐ the labeled magnet used on the van
- Ⓑ the surface used in the investigation
- Ⓒ the position of the magnet on the van
- Ⓓ the magnet used on the wooden block

Write your response on the lines provided.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Section 3

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section **ONLY**. Do not go past the stop sign.

Read all of the information. Use the information to answer the questions.

Simple Cardboard Invention May Help Billions

Not every useful invention has to be developed over many years and cost a lot of money to make. Sometimes the simplest of ideas can improve our lives profoundly. All it takes is a little imagination and a desire to make a difference!

Ask John Bohmer, whose weekend project with his two daughters may end up not only make a significant difference in helping our environment but also improving the lives of 3 billion people worldwide.

John's invention is a solar-powered oven made from cardboard. Dubbed¹ Kyoto Box, it is composed of two cardboard boxes, one inside the other, covered with a sheet of acrylic glass—to help trap the heat from the sun. The inner box is painted with black paint, which absorbs the heat, while the box on the outside is lined with silver foil that reflects any energy that escapes toward the black box. This helps to create a concentrated source of energy that can be used for cooking.

The simple oven costs only \$7 USD to manufacture² and can easily be put together in any cardboard factory. With the capability of reaching high temperatures, it can be used to boil water, cook rice and casserole—and even bake bread.

It is estimated that over 3 billion people living in developing countries use chopped firewood for their main fuel, generating an estimated 2 tons of dangerous carbon dioxide emissions a year.

While solar cookers are not a new idea, this is the first time someone has come up with one that works and is cheap enough to be adapted on a large scale.

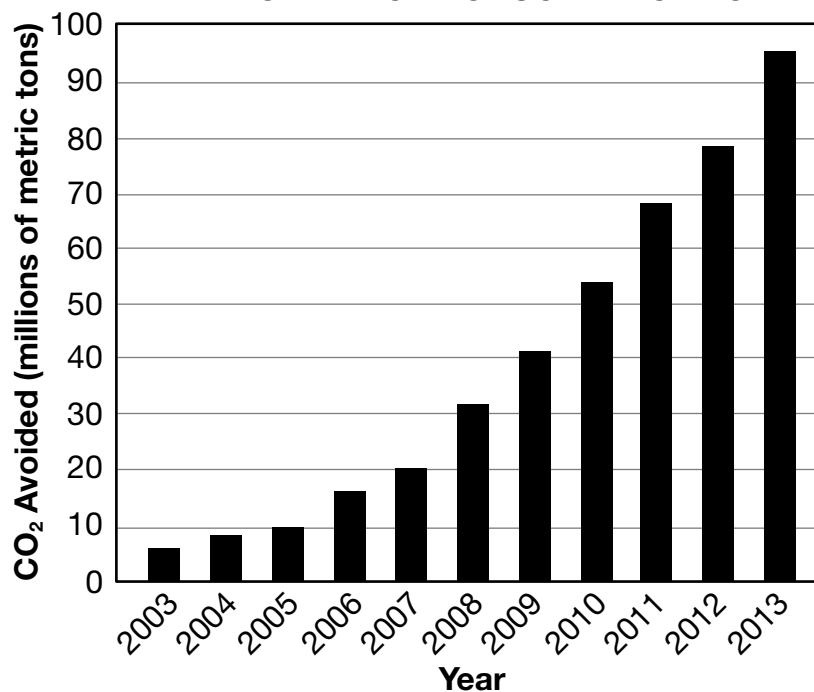
"Simple Cardboard Invention May Help Billions" – Meera Dolasia, DOGO News, © 2009. Reprinted by permission of the publisher

¹**Dubbed:** named

²**manufacture:** make

After reading about the solar-powered ovens, a student was curious about other technologies that help reduce carbon dioxide CO₂ emissions. The student found a credible website that stated that power plants that produce electricity from fossil fuels are a major source of carbon dioxide emissions. Using wind turbines to produce electricity lessens the amount of electricity needed from plants powered by fossil fuels. The use of an average-sized wind turbine to produce electricity can avoid the production of over 3,300 metric tons of carbon dioxide each year. That is equal to the amount of carbon dioxide emitted from 500 cars in one year. The graph shows that from 2003 through 2013 the increasing use of electricity produced by wind power has reduced carbon dioxide emissions by an amount equal to taking 16.9 million cars off of the road.

AMOUNT OF CO₂ EMISSIONS AVOIDED BY USING WIND POWER TO PRODUCE ELECTRICITY



Producing electricity from wind turbines instead of at plants powered by fossil fuels seems likely to help reduce the amount of carbon dioxide in the air. However, while wind turbines do not produce carbon dioxide, they do require the use of large areas of land.

1 John Bohmer's invention is inexpensive technology that uses

- Ⓐ renewable resources to allow food to last longer.
- Ⓑ renewable resources to heat food before it is eaten.
- Ⓒ nonrenewable resources to protect food from germs.
- Ⓓ nonrenewable resources to make appliances more affordable.

2 Which environmental impacts would occur if John Bohmer's invention were used more often?

Select three.

- Ⓐ an increased use of fossil fuels
- Ⓑ a decreased use of fossil fuels
- Ⓒ more use of renewable resources as energy sources
- Ⓓ more use of nonrenewable resources as energy sources
- Ⓔ an increased amount of carbon dioxide in the atmosphere
- Ⓕ a decreased amount of carbon dioxide in the atmosphere

3 The increased use of the invention would most likely increase

- Ⓐ the need to chop down trees.
- Ⓑ the need to dig for fossil fuels.
- Ⓒ the use of nonrenewable resources.
- Ⓓ the quality of the air in the atmosphere.

- The effect of using 5 wind turbines to generate electricity for one year is equal to reducing how many metric tons of carbon dioxide emitted by how many cars?**

- GO ON ►**

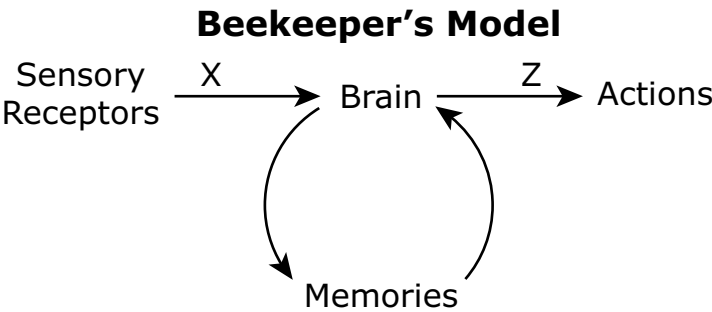
Write your response on the lines provided.

[illegible]

Read all of the information. Use the information to answer the questions.

Students at recess kicked a ball that hit a tree. The students saw a swarm of bees fly out of the tree. After school, the teacher asked a beekeeper to move the bees to a safe place. After moving the bees, the beekeeper talked to the students. The students learned how honeybees sense their environment.

The beekeeper drew a model. The model shows how the bees’ sensory receptors are connected to the bees’ actions.



Next, the beekeeper showed a table about some of the bees’ sensory receptors.

Receptor Location	Detects
Antenna	Smell, Taste, Touch, Sound (vibrations)
Tongue	Taste
Eye	Light
Body Hair	Vibrations, Touch

Beekeeper's Chart of Bee Facts

1. The first step in the process is to identify the problem or issue that needs to be addressed. This involves gathering information and understanding the context of the problem.

- 7 In the Beekeeper's Model, what does arrow "X" represent?**
- Ⓐ energy sent from the environment to the brain
 - Ⓑ information sent from a receptor to the brain
 - Ⓒ information sent to a muscle from the brain
 - Ⓓ matter sent to a receptor from the brain
- 8 Which statement best describes the Beekeeper's Model?**
- Ⓐ It represents actions that bees can take without using memories.
 - Ⓑ It represents the body parts that interact with the environment.
 - Ⓒ It represents a system that allows bees to react as a group.
 - Ⓓ It represents a system that allows the bees to take actions.
- 9 Which two statements describe the information that is transmitted through the waggle dance?**
- Ⓐ It causes bees to react without forming a memory.
 - Ⓑ It communicates the distance and direction of the flowers.
 - Ⓒ It is stored as memories by the bees that found the flowers.
 - Ⓓ It causes bees to immediately leave the hive to attack a predator.
 - Ⓔ It is stored in the sensory receptors activated by the waggle dance.

Write your response on the lines provided.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Section 4

Directions:

Today, you will take Section 4 of the MISA Practice Test.

Read each question. Then, follow the directions to answer each question. Mark your answers by completely filling in the circles in your test book. Do not make any pencil marks outside of the circles. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the lined space provided in your test book. Be sure to keep your response within the provided space. Only responses written within the lined box provided will be scored.

- If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this Section ONLY. Do not go past the stop sign.

Read all of the information. Use the information to answer the questions.

Two students are working together to observe a local ecosystem as part of a school project. The students decide to use their backyards because one has a sunflower garden and the other has a small pond.

The students made observations during the spring, summer, and early fall. In their first observation, the students found that green frogs were living in the pond. They also found that swallowtail butterflies were living in the garden.

The students focused their observations on the green frogs, the swallowtail, and the sunflower. They created the following tables to summarize what they observed.

Green Frog Observations

When	Observation
May 1	Found adult frogs near the pond
May 15	Pond now has several large groups of eggs
May 20	Pond has small tadpoles that have only a head and tail
August 10	Tadpoles now have legs and shorter tails
September 30	Young frogs with four legs and no tails are found outside of the pond

Swallowtail Observations

When	Observation
June 16	Found eggs on leaves
June 25	Plants now have small black caterpillars living on them
July 15	The caterpillars have become larger
July 20	Caterpillars have attached to leaves and are hanging upside down
July 21	Chrysalis forms around the hanging caterpillar
August 2	New adult butterfly breaks out of the chrysalis and flies away

Sunflower Plant Observations

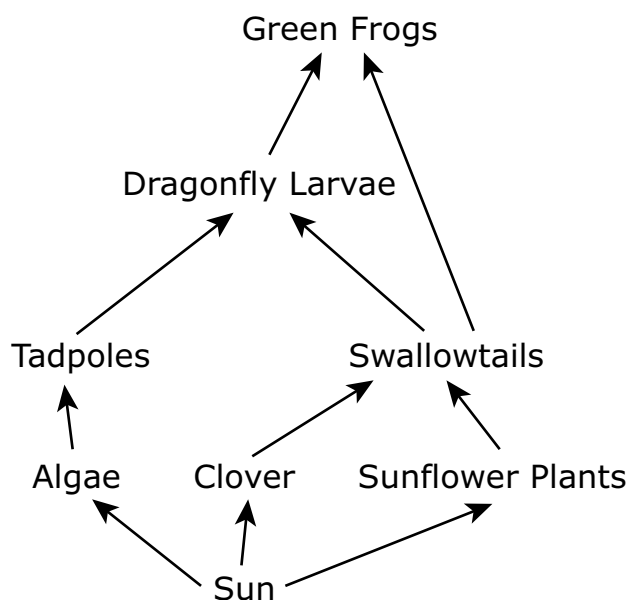
When	Observation
May 15	Planted sunflower seeds
May 19	Found a seed on top of the soil that was split open with a small root growing
May 23	A tiny stem with small leaves appears above soil
June 30	Stem grows taller and many leaves develop
August 7	Flower bud develops at the tip of the stem
August 31	The bud opens into a flower
September 15	Seeds develop in the flower
October 1	Seeds drop to the soil

The students also recorded the following general observations.

1. There were more seeds or eggs than young plants or animals.
2. There were more young plants or animals than there were adult plants or animals.
3. Sunflower plants bend so that leaves face the sun during the day.
4. Some caterpillars and other insects eat the sunflower leaves.
5. Some butterflies land on the flowers of the sunflower.

The students created a diagram to show some of the energy transfers observed in their backyard.

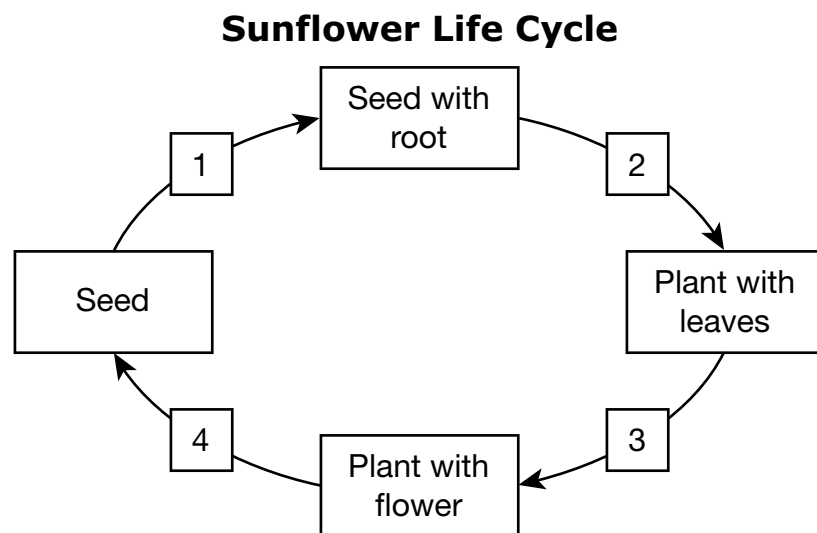
Backyard Energy Transfers



1 Based on the students' observations, what is common among the three organisms?

- Ⓐ Organisms start small and become larger over time but keep the same shapes.
- Ⓑ Organisms remain in the same environment where they were born.
- Ⓒ Organisms have the same food requirements during their life cycles.
- Ⓓ Organisms are born, grow, reproduce, and die.

2 A model shows the life cycle of a sunflower.



Which arrow best represents the process of reproduction?

- Ⓐ 1
Ⓑ 2
Ⓒ 3
Ⓓ 4

- 3 Which statement best explains why sunflower stems bend so that the leaves face the sun during the day?**
- Ⓐ to absorb heat to stay warm during the cool day
 - Ⓑ to receive the most light energy for growth
 - Ⓒ to prevent caterpillars from feeding on the warm leaves
 - Ⓓ to attract pollinators to the light reflected by the flower
- 4 Based on the observations, which statement best identifies how the tadpole uses the energy it gets from eating?**
- Ⓐ to hunt insects
 - Ⓑ for reproduction
 - Ⓒ to hatch from eggs
 - Ⓓ for growing into an adult frog
- 5 Which statement best describes the energy shown in the Backyard Energy Transfers diagram?**
- Ⓐ The producers make energy using air and water.
 - Ⓑ Consumers use all the available energy for movement.
 - Ⓒ The energy in the food of dragonflies was once from the sun.
 - Ⓓ Green frogs receive more energy from the sun than other organisms do.

6 Use the Backyard Energy Transfers diagram to describe how tadpoles receive the energy required to grow into adult frogs.

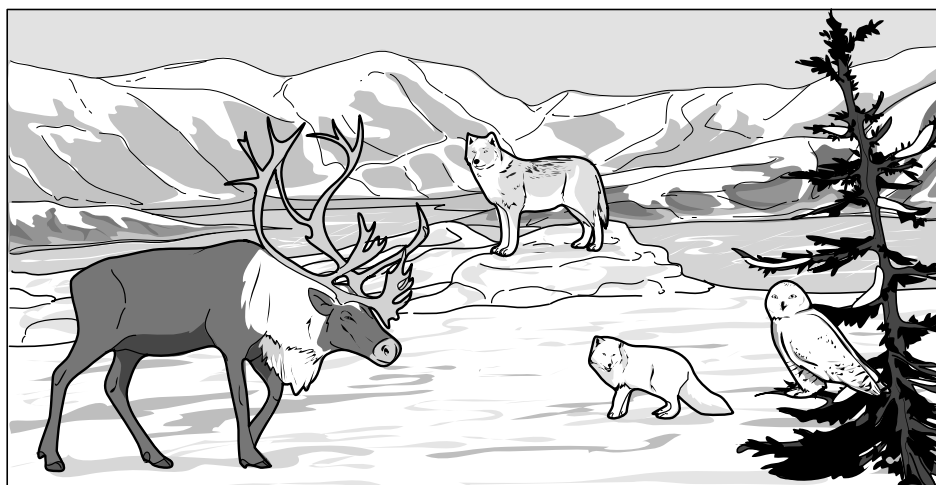
Write your response on the lines provided.

[illegible]

Read all of the information. Use the information to answer the questions.

During a unit on the environment, students researched ecosystems such as rain forests, deserts, and tundras. A group of students researched tundras and the many animals that live in a tundra. The students questioned which characteristics of the animals in the tundra help them to survive.

The students chose four animals that are found in a tundra environment and researched information about each, shown in the table.



61

PLEASE DO NOT WRITE IN THIS AREA

SERIAL #

While researching tundra animals, the students found that the animals' environment changes. Higher temperatures have reduced the amount of snow present in the tundra, leading to more springlike and summerlike days. The students investigated how changes to the environment may affect the animals that live in a tundra.

To observe the changes to the environment and how the animals would appear during each season, the students constructed four diagrams, as shown.

The research indicated some tundra animals live in large groups while others live alone or in small groups. The research also indicated that prey animals tend to live in larger groups and predators live in smaller groups.

The students then constructed diagrams of the Arctic Fox, Arctic Wolf, Caribou, and Musk Oxen to observe whether the animals live alone, in small groups or in large groups, shown as follows.



PLEASE DO NOT WRITE IN THIS AREA

SERIAL #

- 7 The students stated that even though the organisms have different characteristics that enable them to survive in a tundra environment, all the organisms have basic needs.**

Determine the basic needs that organisms require.

Select three.

- Ⓐ feathers
- Ⓑ food
- Ⓒ fur
- Ⓓ shelter
- Ⓔ water

- 8 The students claimed that the arctic fox has a characteristic to help it survive from winter to summer.**

The characteristic that best helps the arctic fox survive is the ability to

- Ⓐ be more visible to prey.
- Ⓑ find more water sources.
- Ⓒ be less visible to predators.
- Ⓓ find fewer burrows for shelter.

- Select the two animals that would most likely be able to survive in a tundra environment.



Snake



Polar Bear



Oriole Bird



Turtle



Toucan Bird



Husky Dog

- 10 The students claimed that animals that live in smaller groups need to develop other characteristics in order to survive in their environment.**

The animals that use blending into the environment instead of living in large groups as a way of survival are

- Ⓐ caribou and arctic fox.
 - Ⓑ snowy owl and caribou.
 - Ⓒ musk ox and arctic wolf.
 - Ⓓ arctic fox and arctic wolf.
- 11 The circling behavior of the musk oxen most likely benefits the herd by**
- Ⓐ providing more food for the herd.
 - Ⓑ finding better shelter for the herd.
 - Ⓒ reducing the number of young in the herd.
 - Ⓓ protecting the smaller animals in the herd.

- 12 Use evidence from the research to support the claim that being part of a group helps the animals survive.**

Write your response on the lines provided.

Handwriting practice area with 20 horizontal lines.



PLEASE DO NOT WRITE IN THIS AREA

Serial number input area with 20 circles, the first of which is checked.

SERIAL #

The following pages include the answer key for all machine-scored items. Refer to the separate documents for all hand-scored items as indicated in the answer key.

Section 1

Item Number	Answer Key	Performance Expectation
1.	D	5-LS2-1
2.	D	5-LS2-1
3.	D, E	5-LS2-1
4.	D	5-LS1-1
5.	B	5-LS1-1
6.	Refer to Maryland Integrated Science Assessment Rubrics	5-LS2-1
7.	D	4-PS3-2
8.	B, D, F	4-PS3-4
9.	A	4-PS3-2

Table continues on next page.

Section 1, continued

Item Number	Answer Key	Performance Expectation
10.	B	4-PS3-2
11.	D	4-PS3-2
12.	Refer to Maryland Integrated Science Assessment Rubrics	4-PS3-4

Section 2

Item Number	Answer Key	Performance Expectation
1.	C	5-ESS2-2
2.	B	5-ESS2-1
3.	C	5-ESS2-1
4.	Part A: C Part B: D	5-ESS2-1
5.	B	5-ESS2-2
6.	Refer to Maryland Integrated Science Assessment Rubrics	5-ESS2-2
7.	A, C	3-5-ETS1-3
8.	B	3-PS2-3
9.	D	3-PS2-3
10.	C	3-PS2-3
11.	C	3-5-ETS1-3
12.	Refer to Maryland Integrated Science Assessment Rubrics	3-5-ETS1-3

Section 3

Item Number	Answer Key	Performance Expectation
1.	B	4-ESS3-1
2.	B, C, F	4-ESS3-1
3.	D	4-ESS3-1
4.	B	5-ESS3-1
5.	D	5-ESS3-1
6.	Refer to Maryland Integrated Science Assessment Rubrics	4-ESS3-1
7.	B	4-LS1-2
8.	D	4-LS1-2
9.	B, C	4-LS1-2
10.	B	3-LS2-1
11.	Part A: B Part B: D	3-LS2-1
12.	Refer to Maryland Integrated Science Assessment Rubrics	4-LS1-2

Section 4

Item Number	Answer Key	Performance Expectation
1.	D	3-LS1-1
2.	D	3-LS1-1
3.	B	3-PS3-1
4.	D	3-PS3-1
5.	C	3-PS3-1
6.	Refer to Maryland Integrated Science Assessment Rubrics	3-PS3-1
7.	B, D, E	3-LS4-3
8.	C	3-LS4-3
9.	B, F	3-LS4-3
10.	D	3-LS2-1
11.	D	3-LS2-1
12.	Refer to Maryland Integrated Science Assessment Rubrics	3-LS2-1